PHYS6260 HW2

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1 Data

The data were taken from the Berkeley black hole compilation (found in `current_ascii.txt`) [1]. The fitting variables were defined in log space as

$$x = \log_{10}\left(\frac{\sigma}{200\ \mathrm{km\,s^{-1}}}\right), \qquad y = \log_{10}\left(\frac{M_{\mathrm{bh}}}{M_\odot}\right).$$

The baseline model assumed the usual power-law relation

$$M_{\mathrm{bh}} = A\sigma^b,$$

which becomes linear in the transformed variables:

$$y = \alpha + bx.$$

2 Modeling Approach

For the baseline model, the total uncertainty used in the likelihood was

$$\sigma_{\mathrm{tot},i}^2 = \sigma_{y,i}^2 + b^2\sigma_{x,i}^2,$$

where $\sigma_{y,i}$ and $\sigma_{x,i}$ are the symmetrized log-space uncertainties derived from the catalog's asymmetric linear error bars.

The intrinsic-scatter extension added an extra term ϵ in quadrature:

$$\sigma_{\mathrm{tot},i}^2 = \sigma_{y,i}^2 + b^2\sigma_{x,i}^2 + \epsilon^2.$$

For both runs, 32 walkers and 4000 steps were used, with a burn-in of 1000 steps. The baseline run had a mean acceptance fraction of 0.7130, while the intrinsic-scatter run had a mean acceptance fraction of 0.6416.

3 Posterior Results

3.1 Baseline Model

The baseline posterior highest-posterior-density (HPD) intervals were:

Table 1: Baseline posterior summaries

Parameter	Median	68% HPD	95% HPD
b	5.87171	[5.74844, 5.99043]	[5.63237, 6.11548]
A ($M_\odot$)	6.32876×10^{-6}	[1.80083×10^{-6}, 9.0976×10^{-6}]	[6.69313×10^{-7}, 1.85362×10^{-5}]

The baseline posterior for A and b is shown in Figure 1, and the corresponding median-fit line in log space is shown in Figure 2.

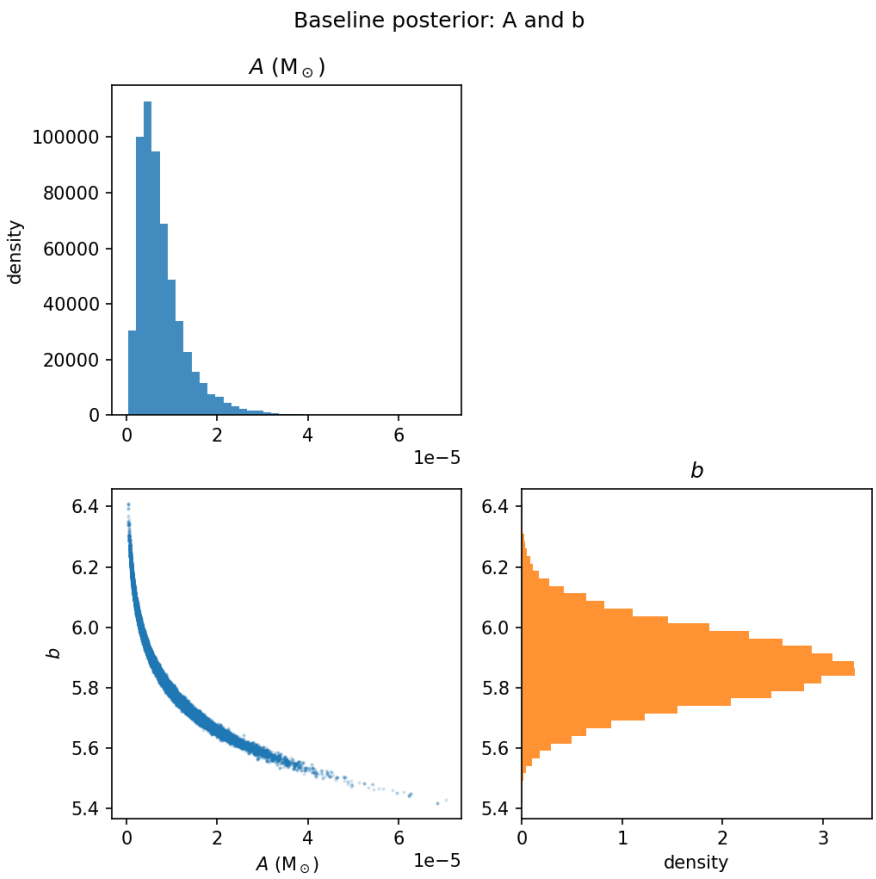

Figure 1: Baseline posterior for A and b.

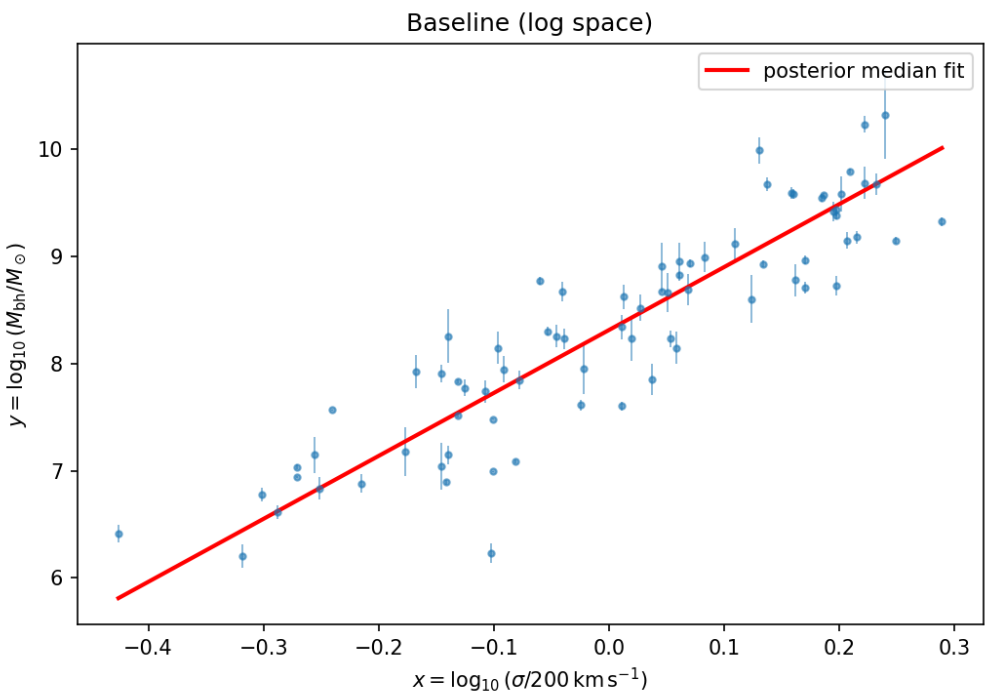

Figure 2: Baseline fit in log space using the posterior median parameters.

3.2 Intrinsic-Scatter Model

For the model including intrinsic scatter, the posterior HPD intervals were:

Table 2: Intrinsic-scatter posterior summaries

Parameter	Median	68% HPD	95% HPD
b	5.39532	[5.09968, 5.67894]	[4.79859, 5.96236]
A ($M_\odot$)	8.02923×10^{-5}	$[9.17099 \times 10^{-8}, 1.67112 \times 10^{-4}]$	$[7.63352 \times 10^{-8}, 1.07602 \times 10^{-3}]$
ϵ (dex)	0.392784	[0.351038, 0.431529]	[0.319383, 0.476900]

The marginal posterior for ϵ is shown in Figure 3.

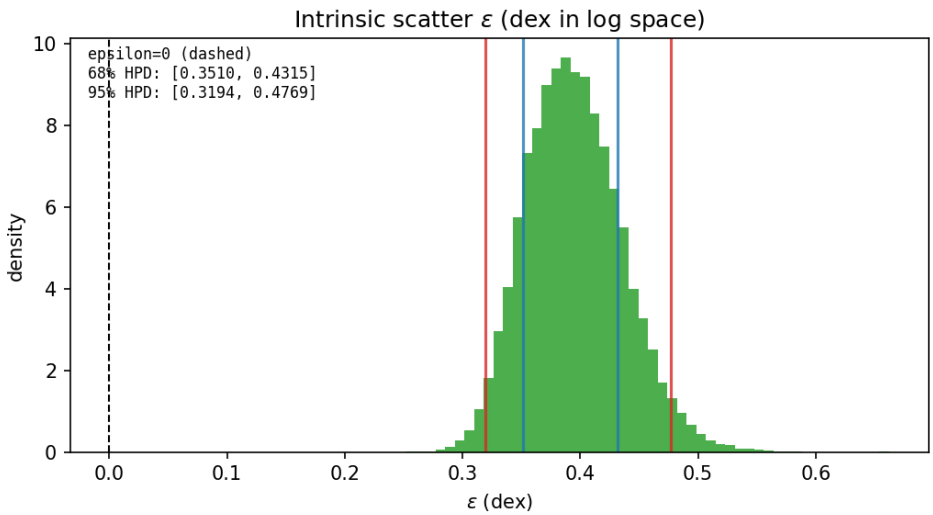

Figure 3: Posterior marginal distribution for the intrinsic scatter parameter ϵ.

3.3 Choices on A and b

The M–σ relation was modeled as

$$M_{\rm bh} = A\sigma^b.$$

Instead of fitting A directly, I rewrote the model in log space as

$$y = \alpha + bx, \qquad y = \log_{10}\left(\frac{M_{\rm bh}}{M_\odot}\right), \qquad x = \log_{10}\left(\frac{\sigma}{200\ {\rm km\,s^{-1}}}\right).$$

In this form, α is just the log black-hole mass when $\sigma = 200\ {\rm km\,s^{-1}}$, and the original normalization can be recovered with

$$A = 10^{\alpha - b\log_{10}(200)}.$$

Going from 10^{-6} to 10^{-5} feels like a similar kind of change as going from 10^{-5} to 10^{-4}, whereas a flat linear prior would put much more weight on the larger values , basically why I think that a log scale makes more sense for A.

I sample α instead of A and using a bounded scale

$$\alpha \sim \text{Uniform}(5, 11).$$

For the slope b, I used a bounded linear prior

$$b \sim \text{Uniform}(0, 10).$$

A linea prior felt more appropriate here because b is not a scale factor like A; it is the exponent that sets how strongly black-hole mass changes with velocity dispersion. Another thins is I also wanted to keep the range physically reasonable and avoid clearly implausible values like large negative slopes, which is what linearity does for us.

4 Consistency of ϵ with Zero and Intrinsic Scatter

To check whether the M–σ relation needs sum extra scatter beyond the quoted measurement errors, I extended the baseline mode by adding an intrinsic-scatter term ϵ in log space:

$$\sigma_{\text{tot},i}^2 = \sigma_{y,i}^2 + b^2\sigma_{x,i}^2 + \epsilon^2.$$

In this setup, ϵ represents additional Gaussian scatter in

$$y = \log_{10}\left(\frac{M_{\text{bh}}}{M_\odot}\right)$$

at fixed σ, on top of the propagated observational uncertainties.

From the intrinsic-scatter MCMC run, the posterior median came out to

$$\epsilon = 0.392784 \text{ dex},$$

and the 95% HPD interval was

$$\epsilon \in [0.319383,\ 0.476900] \text{ dex}.$$

Because the lower bound of this interval is clearly > 0, ϵ isn't consistent with 0 in this model. If it had been consistent with zero, that would mean basically that the spread in the data could be explained just by the measurement uncertainties.

This willl suggest pretty much that the relation is real and tight, but it is not perfectly noise-free. TLDR there seems to be additional astrophysical variation that is not captured by the simple two-parameter model. This makes sense since this is a linearized model

References

[1] Nicholas J. McConnell and Chung-Pei Ma. "Revisiting the Scaling Relations of Black Hole Masses and Host Galaxy Properties". In: *The Astrophysical Journal* 764.2 (2013), p. 184. DOI: 10.1088/0004-637X/764/2/184. URL: https://blackhole.berkeley.edu/?page_id=68.